Measuring your kinetic energy!

Name: _____

You will be calculating the kinetic energy you have when you run across the basketball court.

Step one: measure the length of the basketball court (in metres) using a trundle wheel.

LENGTH OF BASKETBALL COURT= _____m

Step two: measure time it takes (in seconds) to run from one end of the basketball court to the other using a timer. Do this 3 times to get an average

First Run	Second Run	Third Run	Average

TIME TAKEN TO RUN LENGTH OF BASKETBALL COURT = _____s

Step three: work out your speed in metres per second

$\frac{distance \in m}{time \in seconds}$ = speed in meters per second (m/s)

SPEED= _____ m/sec

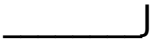
Step four: measure your mass using the bathroom scales.

MASS = _____kg

Step five: work out your kinetic energy in joules.

Kinetic energy = ½ x objects mass x (objects speed)^2 *or*
$E_k = ½ ms^2$

Kinetic energy = ½ x _____ x (_____m/s)^2

= ½ x _____

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